

E-Commerce Sales Analysis Dashboard Using Power BI

Project Overview

Objective

The objective of this project is to analyze e-commerce sales performance using Power BI by creating calculated columns, DAX measures, and interactive visualizations. The dashboard helps stakeholders monitor sales performance, compare actual sales against targets, identify profitable product categories, and analyze regional sales trends.

Data Model

Tables Used

- 1. Orders
- 2. Order Details
- 3. Sales Target
- 4. Date Table

Data Relationships

☐ From: table (column) ↑	Relationship	To: table (column)	Status	
☐ List of Orders (Order ID)	1 — 1	Order Details (Order ID)	Active	...
☐ Order Details (Category)	* — *	Sales target (Category)	Active	...
☐  Order Details (order date)	* — 1	Table (Date)	Active	...
☐ Order Details (Order ID)	1 — 1	Orders Data (Order ID)	Active	...
☐ Sales target (Category)	* — 1	Count of orders (Category)	Active	...
☐ Sales target (Month of Order ...)	* — 1	 Orders Data (Order Date)	Inactive	...
☐ Sales target (Month of Order ...)	* — 1	Table (Date)	Active	...
☐ Table (Date)	1 — 1	 Sales target By Month (Month ...)	Active	...

Description:

The data model was created by establishing relationships between Orders, Order Details, Sales Target, and Date tables. This structure enables efficient filtering, aggregation, and time intelligence calculations across the report.

Calculated Columns

1. Category Type

```
1 Category Type = CONCATENATE('Order Details'[Category],CONCATENATE("-",'Order Details'[Sub-Category]))
```

ount ▾	Profit ▾	Quantity ▾	Category ▾	Sub-Category ▾	Profit Margin ▾	Profit Status ▾	Category Type ▾	Revenue_Per_
65	17	2	Clothing	T-shirt	0.261538461538462	Profit	Clothing_T-shirt	
87	4	2	Clothing	Shirt	0.0459770114942529	Profit	Clothing_Shirt	

Purpose:

Combines Category and Sub-Category into a single field to provide more detailed product classification.

Example:

Furniture - Chairs

2. Revenue per Order

```
1 Revenue_Per_Order = 'Order Details'[Amount]*'Order Details'[Quantity]
```

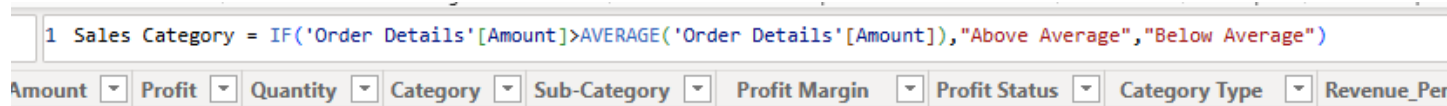
Purpose:

Calculates revenue generated by each order using Amount and Quantity.

Business Value:

Helps analyze order-level revenue contribution.

3. Sales Category



Purpose:

Categorizes orders as Above Average or Below Average based on sales amount.

Business Value:

Helps identify high-performing and low-performing orders.

DAX Measures

1. Order Count

✕ ✓	
Category	Count
Furniture	88
Electronics	95
Clothing	317

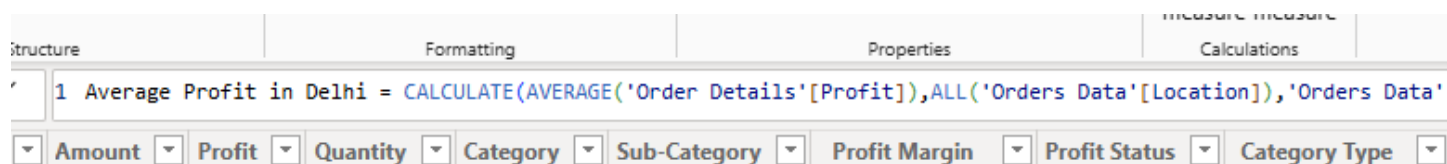
Purpose:

Calculates the total number of orders.

Business Insight:

Provides visibility into overall order volume.

2. Average Profit in Delhi



Purpose:
Calculates average profit for orders placed in Delhi.

Business Insight:
Measures profitability within a specific city.

3. Year-To-Date Sales (YTD)

Structure		Formatting		Properties		Calculations	
1 Total sales Revenue = TOTALYTD(SUM('Order Details'[Revenue_Per_Order]),'Sales target'[Month of Order Date])							
Entity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue_Per_Order	
2	Clothing	T-shirt	0.261538461538462	Profit	Clothing_T-shirt	₹ 130	

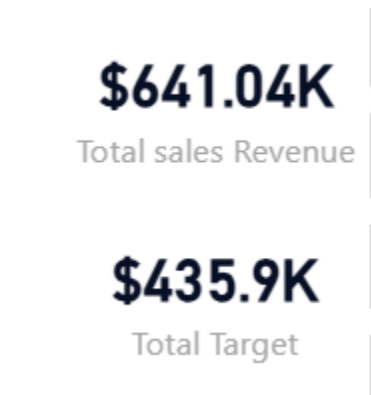
Purpose:
Calculates cumulative sales from the beginning of the year until the selected date.

Business Insight:
Tracks sales growth over time.

Dashboard Visualizations

KPI Cards

Insert Screenshot

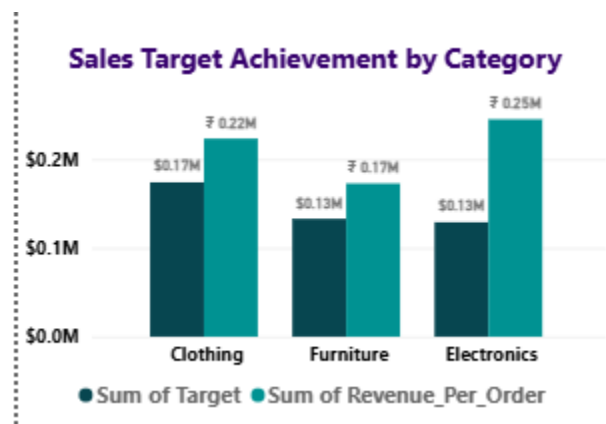


Explanation:
KPI cards provide a quick summary of business performance and allow management to monitor key metrics at a glance.

Sales Target Achievement by Category

Insert Clustered Column Chart Screenshot

Visual Type:
Clustered Column Chart



Explanation:
This visual compares actual sales against sales targets across product categories.

Business Insight:
Identifies categories that are exceeding targets and those requiring attention.

Monthly Sales Trend



Visual Type:

Line Chart

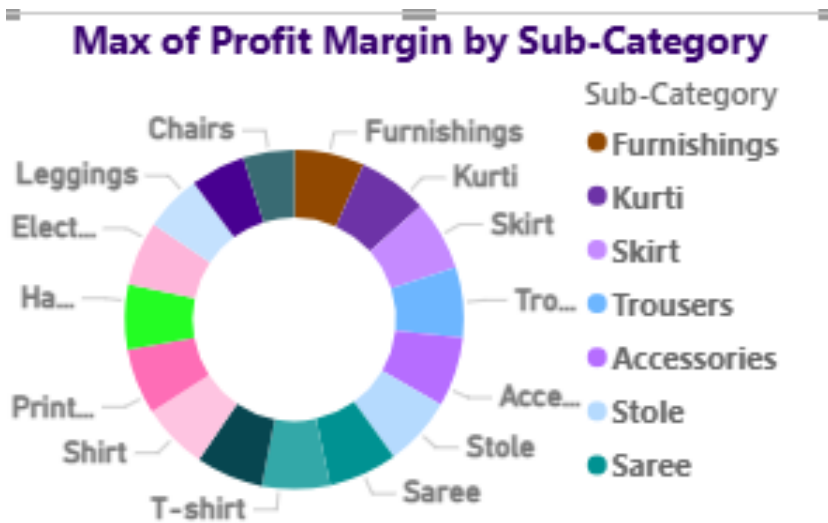
Explanation:

Displays sales performance across months.

Business Insight:

Helps identify seasonal patterns and monthly growth trends.

Maximum Profit Margin by Sub-Category



Visual Type:

Donut Chart

Explanation:

Shows the profit margin contribution of each sub-category.

Business Insight:

Identifies the most profitable product sub-categories.

Sales Distribution by Sub-Category

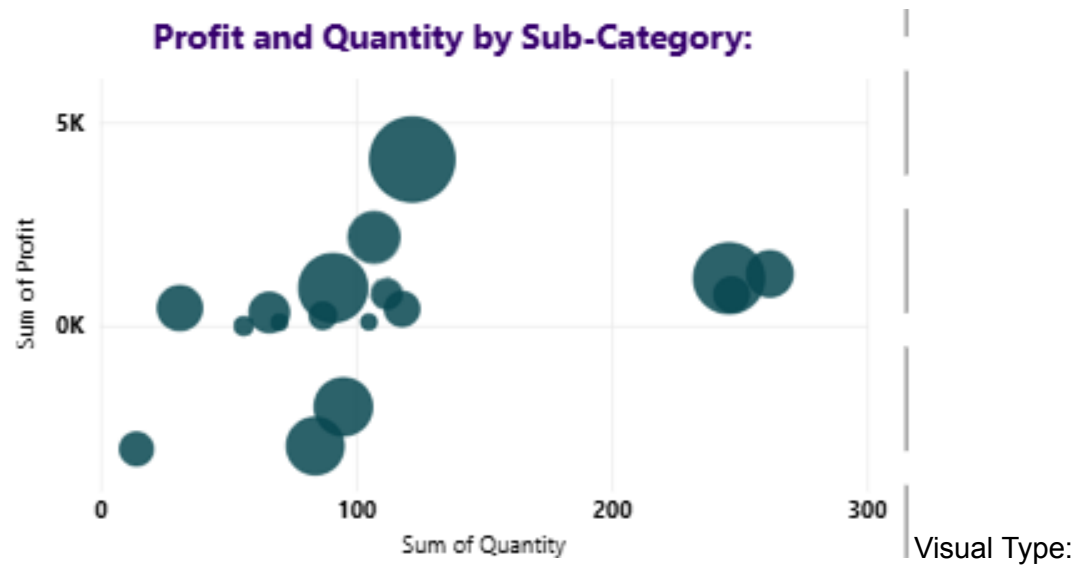


Visual Type:
Treemap

Explanation:
Represents sales contribution across different sub-categories.

Business Insight:
Highlights dominant product segments driving revenue.

Profit vs Quantity Analysis



Scatter Chart

Explanation:
Compares profit and quantity sold for each sub-category.

Business Insight:
Helps determine whether higher sales volume translates into higher profitability.

Sales Performance Matrix

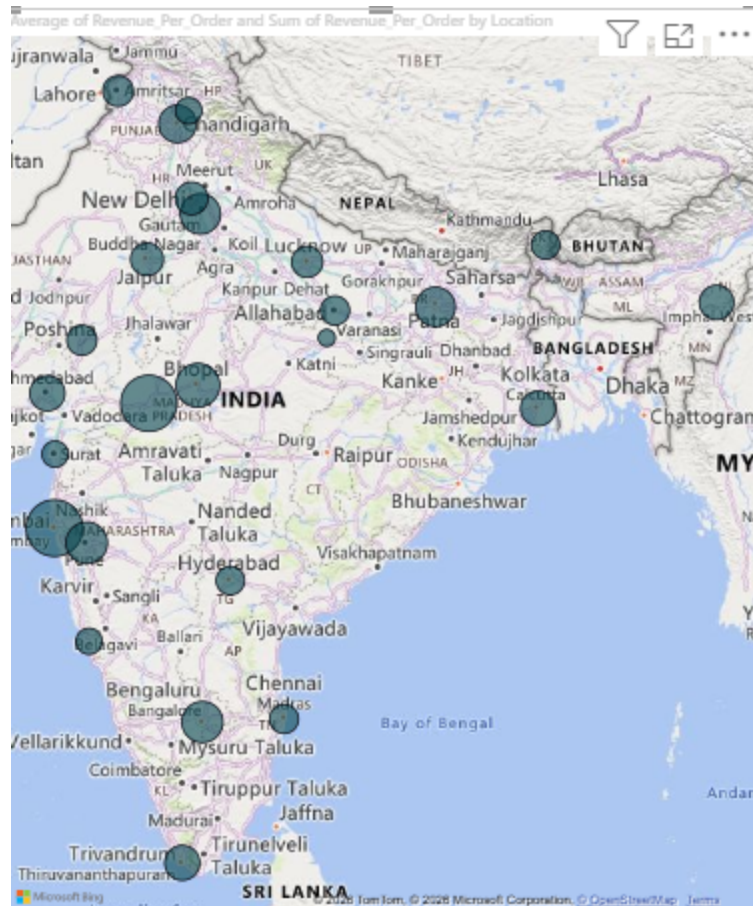
Category Month Target format	Clothing		Electronics		Furniture		Total
	Sum of Reveue	Target Sales Measre	Sum of Reveue	Target Sales Measre	Sum of Reveue	Target Sales Measre	Sum of Reveue
Sep-18	\$58,752	14000	\$33,543	9000	\$40,422	11000	\$1,32,717
Oct-18	\$55,979	16000	\$69,151	9000	\$51,533	11100	\$1,76,663
Nov-18	\$72,687	16000	\$92,824	9000	\$64,274	11300	\$2,29,785
May-18	\$44,338	12000	\$61,765	9000	\$32,664	10500	\$1,38,767
Mar-19	\$92,718	16000	\$98,989	16000	\$1,30,043	11800	\$3,21,750
Jun-18	\$40,292	12000	\$43,234	9000	\$26,489	10600	\$1,10,015
Jul-18	\$13,060	14000	\$22,939	9000	\$17,145	10800	\$53,144
Jan-19	\$66,935	16000	\$1,61,965	16000	\$1,08,329	11500	\$3,37,229
Feb-19	\$49,332	16000	\$48,507	16000	\$79,781	11600	\$1,77,620
Dec-18	\$43,306	16000	\$92,979	9000	\$28,554	11400	\$1,64,839
Aug-18	\$59,474	14000	\$47,050	9000	\$44,990	10900	\$1,51,514
Apr-18	\$67,649	12000	\$43,637	9000	\$41,541	10400	\$1,52,827
Total	\$6,64,522		\$8,16,583		\$6,65,765		\$21,46,870

Visual Type:
Matrix

Explanation:
Displays actual sales and targets across categories and months.

Business Insight:
Provides detailed performance comparison for business monitoring.

Geographic Sales Analysis



Visual Type:

Map

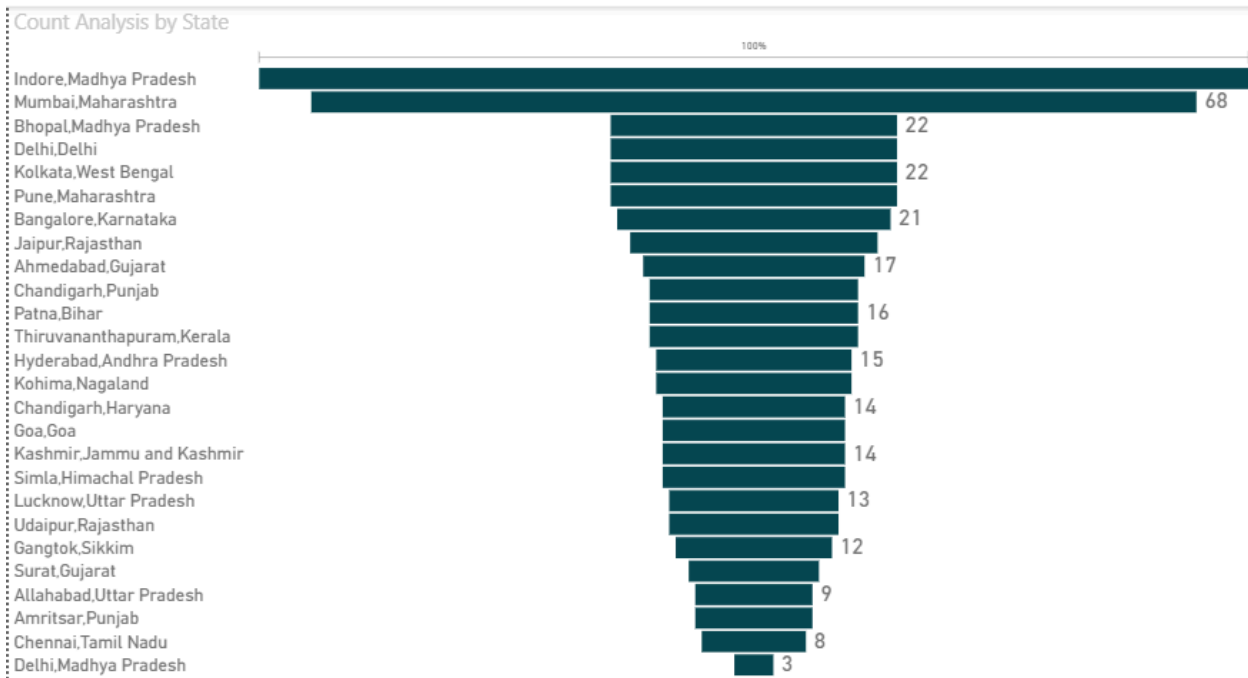
Explanation:

Visualizes sales across cities.

Business Insight:

Identifies high-performing and low-performing regions.

Order Count by State



Visual Type:
Funnel Chart

Explanation:
Displays the distribution of orders across states.

Business Insight:
Highlights states contributing the highest number of orders.

Minimum Target by Segment

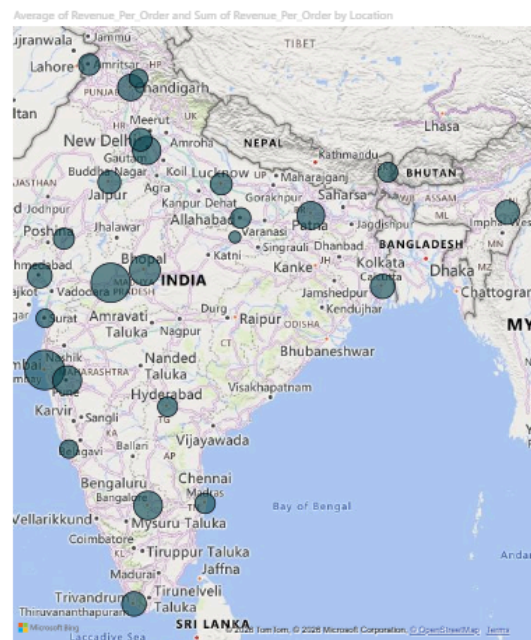
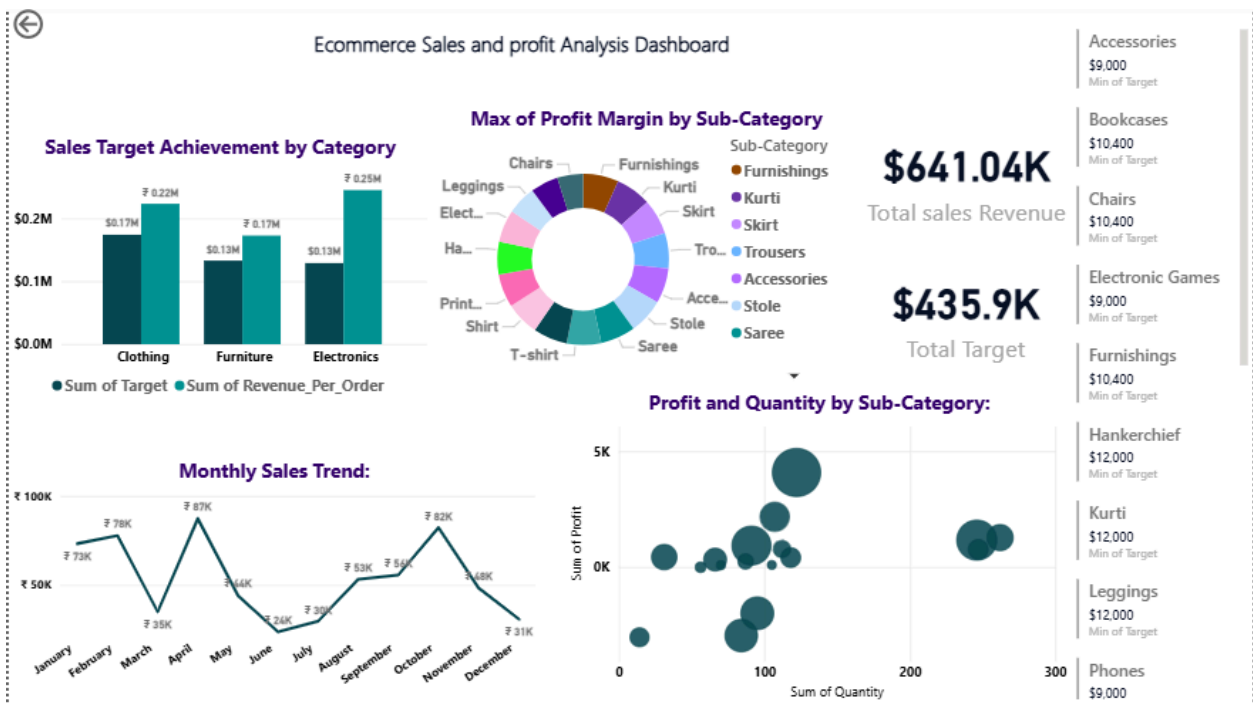


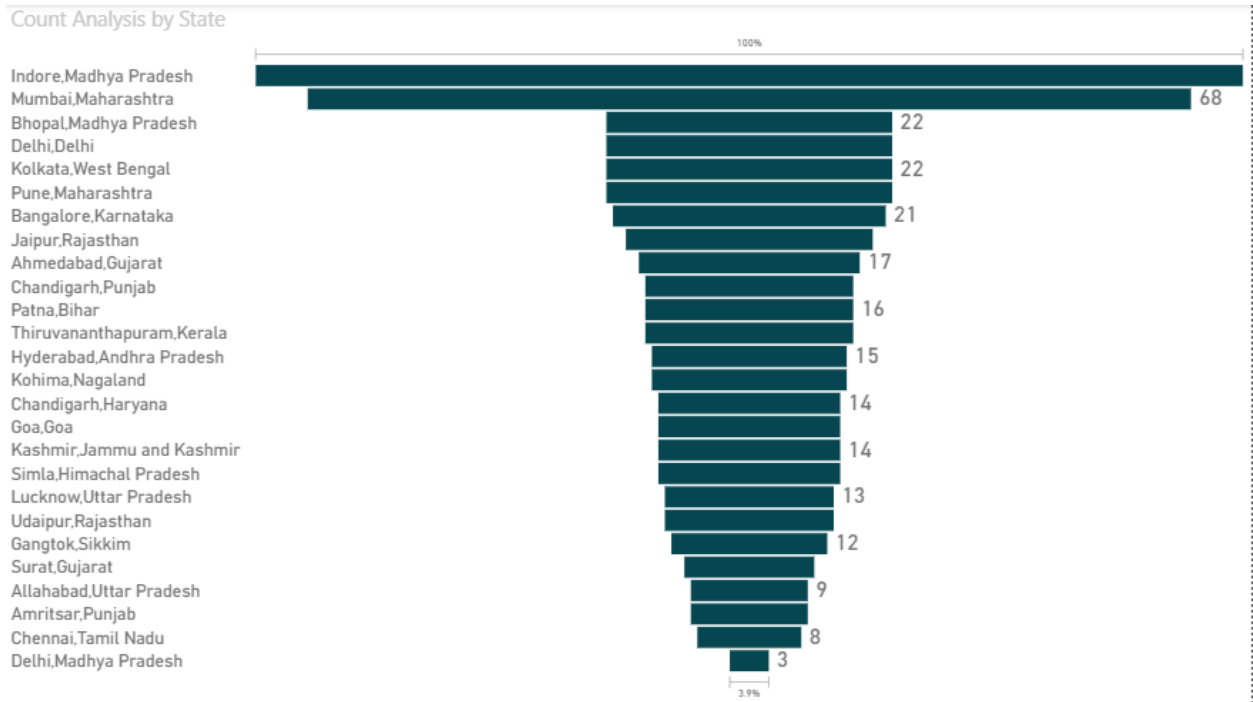
Visual Type:
Multi-row Card

Explanation:
Shows the minimum target assigned to each segment.

Business Insight:
Helps compare target expectations across customer segments.

Dashboard Layout





Description:

The dashboard was designed using a clean and structured layout. KPI cards are placed at the top for quick performance monitoring, while trend, comparison, and geographic visuals are positioned below to support detailed analysis and decision-making.

Business Analysis

1. Descriptive Analysis (What Happened?)

Descriptive analysis focuses on summarizing historical sales performance using the available data.

Findings

- Total Sales Revenue achieved was ₹641.04K.
- Total Sales Target was ₹435.9K.
- Actual sales exceeded the assigned target.
- Electronics generated the highest sales among all categories.
- Bookcases, Sarees, and Printers were among the highest revenue-generating sub-categories.
- Monthly sales peaked during April and October.

- Indore and Mumbai recorded the highest number of orders.

Conclusion

The business demonstrated strong sales performance overall, with several product categories exceeding their sales targets.

2. Diagnostic Analysis (Why Did It Happen?)

Diagnostic analysis investigates the reasons behind business performance.

Findings

- Electronics outperformed other categories due to higher customer demand and larger sales volumes.
- Certain sub-categories generated higher profits despite lower sales quantities, indicating stronger profit margins.
- Seasonal spikes observed in April and October suggest increased customer purchasing activity during those periods.
- Geographic analysis indicates stronger market penetration in cities such as Indore and Mumbai.
- Categories with lower sales may require additional promotional efforts or inventory optimization.

Conclusion

Sales performance was influenced by product demand, profit margins, seasonal buying patterns, and regional customer concentration.

3. Predictive Analysis (What Is Likely to Happen?)

Predictive analysis uses current trends to estimate future business outcomes.

Predictions

- If current growth trends continue, total sales are expected to remain above target levels in future periods.
- Electronics is likely to continue being the highest-performing category.

- High-performing sub-categories such as Bookcases and Sarees are expected to contribute significantly to future revenue.
- Cities with high order volumes are likely to remain major revenue contributors.
- Seasonal peaks may continue to occur around months that historically show higher sales activity.

Expected Business Impact

Future sales growth can be achieved by focusing resources on top-performing categories and regions.

4. Prescriptive Analysis (What Should Be Done?)

Prescriptive analysis recommends actions to improve business performance.

Recommendations

1. Increase inventory availability for high-demand products in Electronics and top-performing sub-categories.
2. Allocate additional marketing budget to high-performing cities such as Indore and Mumbai.
3. Introduce promotional campaigns for low-performing categories to improve sales performance.
4. Focus on products with high profit margins rather than only high sales volumes.
5. Prepare inventory and marketing campaigns before peak sales months to maximize revenue opportunities.
6. Regularly monitor sales targets and category performance through the Power BI dashboard.

Expected Outcome

Implementing these recommendations can improve revenue growth, increase profitability, and support better business decision-making.

Overall Business Summary

The analysis reveals that the company is performing above its sales targets, with Electronics leading category performance and several sub-categories contributing significantly to revenue.

Regional sales are concentrated in a few major cities, while seasonal trends influence monthly performance. By focusing on high-performing products, profitable categories, and strategic geographic markets, the organization can further enhance sales growth and profitability.